IT367 – Software Project Management

LAB 4

Software Size Estimation for Abhinay

Autumn 2026–27 | IIIT Vadodara

A. Group Information

Group No.	1	**Project Title**	Abhinay
Date	01-09-2026	**Project Manager**	Ishan Agrawal
Estimation & Cost Manager	Yuvaraju	**Client Representative(s)**	Arjun S

B. Reference from Earlier Labs

Earlier Output	What you will use today
Lab 2 – Requirements & Scope	The approved functional requirements and in-scope features of Abhinay: **user registration/authentication, professional profiles, casting role posting, role discovery and applications, shortlist management, applicant review, application status tracking, direct messaging, audition/screening, self-tape submission, project pages, crew profile linking, social networking, and admin/moderation.**
Lab 3 – WBS	The **14 Level-3 WBS work packages** are used as the individual components for software size estimation: **1.1.1 User Registration & Authentication, 1.1.2 Professional Profile Management, 1.2.1 Casting Role Posting, 1.2.2 Role Discovery & Application, 1.2.3 Shortlist Management, 1.3.1 Applicant Review Dashboard, 1.3.2 Application Status Tracking, 1.4.1 Direct Messaging, 1.4.2 Scheduled Audition/Screening, 1.4.3 Self-Tape Submission, 1.5.1 Project Page Management, 1.5.2 Crew Profile Linking, 1.5.3 Social Networking Feed, and 1.6.1 Admin Dashboard.**

C. Choose One Simple Size Measure

Select	Size Measure	Simple Meaning
☐	LOC	Estimated Lines of Code
☐	Feature Count	Number of major software features/functions
☐	Screen / Report Count	Number of screens, forms or reports
☐	Other (faculty approved)	Write the measure and counting rule below

Selected measure: LOC Unit: Lines of Code

Counting rule / basis: Estimate application-owned frontend, backend, validation, business-logic, database-access and integration code; exclude framework/third-party source, node_modules, generated Prisma client/code, and documentation.

D. Component-wise Software Size Estimation

No.	WBS Code	Component / Feature	Estimated Size	Basis / Assumption	Confidence
1	1.1.1	User Registration & Authentication	400 LOC	Registration for six roles, login/logout, password hashing, JWT access/refresh tokens, RBAC, validation and auth middleware.	H
2	1.1.2	Professional Profile Management	500 LOC	Profile CRUD, skills, experience, bio, CV/resume, profile photo and reel/portfolio upload handling.	M
3	1.2.1	Casting Role Posting	300 LOC	Create, edit, publish and close casting roles with description, requirements, compensation and location.	H
4	1.2.2	Role Discovery & Application	400 LOC	Browse/search/filter casting roles, view details, submit applications and validate application rules.	M
5	1.2.3	Shortlist Management	200 LOC	Create/manage named shortlist folders and add/remove applicants from existing application data.	H
6	1.3.1	Applicant Review Dashboard	250 LOC	Producer/director dashboard to view applicants for each posting, with profile/application details.	M
7	1.3.2	Application Status Tracking	300 LOC	Applied/Shortlisted/Selected/Rejected states, status validation/history and applicant notification on change.	M
8	1.4.1	Direct Messaging	450 LOC	Conversation/message APIs, inbox UI, persistent message state and real-time exchange using a communication layer.	M
9	1.4.2	Scheduled Audition / Screening	250 LOC	Schedule/update/cancel audition sessions, participant/session data and notifications; excludes live video conferencing unless scope is changed.	L
10	1.4.3	Self-Tape Submission	350 LOC	Video upload, application association, file validation, secure media-storage integration and playback/access handling.	L
11	1.5.1	Project Page Management	300 LOC	Create/edit/view film or show project pages and attach cast/crew information.	M
12	1.5.2	Crew Profile Linking	150 LOC	Link crew profiles to projects and show associated projects on professional profiles.	H
13	1.5.3	Social Networking Feed	450 LOC	Connect/follow functionality, create/view industry/project posts and basic feed retrieval.	M
14	1.6.1	Admin Dashboard	450 LOC	Admin moderation of users/posts, credit verification and report handling with role protection.	M

E. Total Estimated Software Size

Selected size measure	LOC
Total estimated size (sum of Section D)	4750 LOC
Unit	Lines of Code
What does this total mean for your project?	Approximately 4,750 LOC of application-owned code is expected for the current approved in-scope of functionality. This is a planning estimate, not an exact final code count.

F. Estimation Assumptions

No.	Assumption
1	Each Level 3 WBS work package is estimated independently based on its expected implementation complexity.
2	The estimate includes application-owned frontend, backend, validation, business-logic, database-access and integration code, while third-party library/framework code is excluded.
3	External services required for media storage and real-time communication are assumed to provide usable

	APIs/SDKs; custom infrastructure development is not included.
4	Only the approved in-scope functionality is included in the estimate; advanced out-of-scope features such as premium subscriptions, script breakdown, union/guild verification and advanced video-conferencing auditions are excluded.
5	The LOC values are approximate estimates based on feature complexity and may change if requirements, implementation of technology or project scope changes.

G. Uncertainty Check

Which component has the highest uncertainty?	1.4.2 Scheduled Audition / Screening
Why is this estimate uncertain?	The SRS mentions low-latency live video, while the Lab 3 scope of clarification excludes video-conferencing-based auditions. The final interpretation directly changes the implementation size and technical complexity.
What information would help you improve this estimate?	Client confirmation on whether live video is required, the exact audition workflow, notification method, external communication service, session limits and media/streaming requirements.
If the project scope changes, what should you do with this estimate?	Re-estimate the affected WBS component(s), update the total software size, record the change, and use the revised baseline for later effort, cost, and schedule planning.

H. Quick Consistency Review

Check	Yes/No	Correction / Comment
Same project and approved scope from earlier labs used.	Yes	Based on Abhinay requirements and scope from Lab 2.
Major components match Lab 3 WBS.	Yes	All 14 Level 3 WBS work packages are included.
Same size measure/counting rule used for all components.	Yes	LOC is used consistently for all components.
Every important estimate has a reason/assumption.	Yes	Basis/assumptions are provided for each component.
Total size is correctly calculated.	Yes	Total estimated size = **4,750 LOC**.

I. Final Reflection

This lab helped us estimate the approximate software size of the Abhinay platform as 4,750 LOC across its 14 Level-3 WBS components. The component-wise estimates show which features are relatively larger or more uncertain and provide a useful basis for estimating development effort, cost and schedule in later project-management activities.

J. Submission Checklist

☐ Same project from Labs 1–3 used.
☐ One size measure selected and stated clearly.
☐ Component-wise size estimation completed.
☐ Total estimated software size calculated.
☐ Basis/assumptions written clearly.
☐ Highest-uncertainty component identified.
☐ Project Manager and Estimation & Cost Manager reviewed the document.

☐ One completed document is being submitted per group.

K. Sign-off

Project Manager	Estimation & Cost Manager	Client Representative
Name / Signature: Ishan Agrawal	Name / Signature: Yuvaraju	Name / Signature: Arjun S

Keep this Lab 4 output safely. It will be used as an input for later estimation and planning activities.